

THREAT-INTELLIGENCE KILL-CHAIN CORRELATION ENGINE



Correlating Security Alerts into Attack Sequences Using Graph-Based Algorithms

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INTRODUCTION



THE CHALLENGE

Large volumes
of security logs

Manual analysis
is slow

Individual alerts may not
reveal the full attack

THE IDEA

Detect known
attack patterns

Connect
related alerts

Reconstruct
the attack sequence

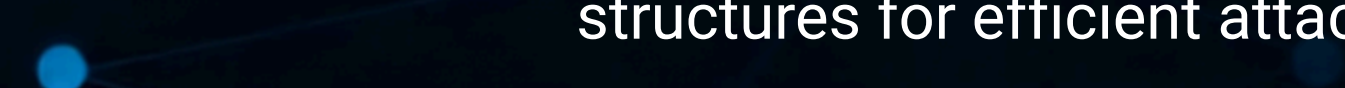
THE GOAL

Analyse correlated
attack paths

Identify
critical points

Prioritise
important alerts

DSA Perspective: Security alerts are transformed into searchable patterns, graphs and priority-based structures for efficient attack analysis



ABSTRACT

- SOC's receive large volumes of security logs and alerts, making manual analysis slow and difficult.
- The proposed system automatically identifies known attack patterns and correlates related alerts.
- Correlated alerts are represented as a graph to reconstruct multi-stage cyberattack sequences.
- Hashing, pattern matching, BFS/DFS, graph optimization and priority queues are used for efficient analysis and alert prioritization.
- The system helps identify critical attack paths, monitoring points and high-priority alerts.

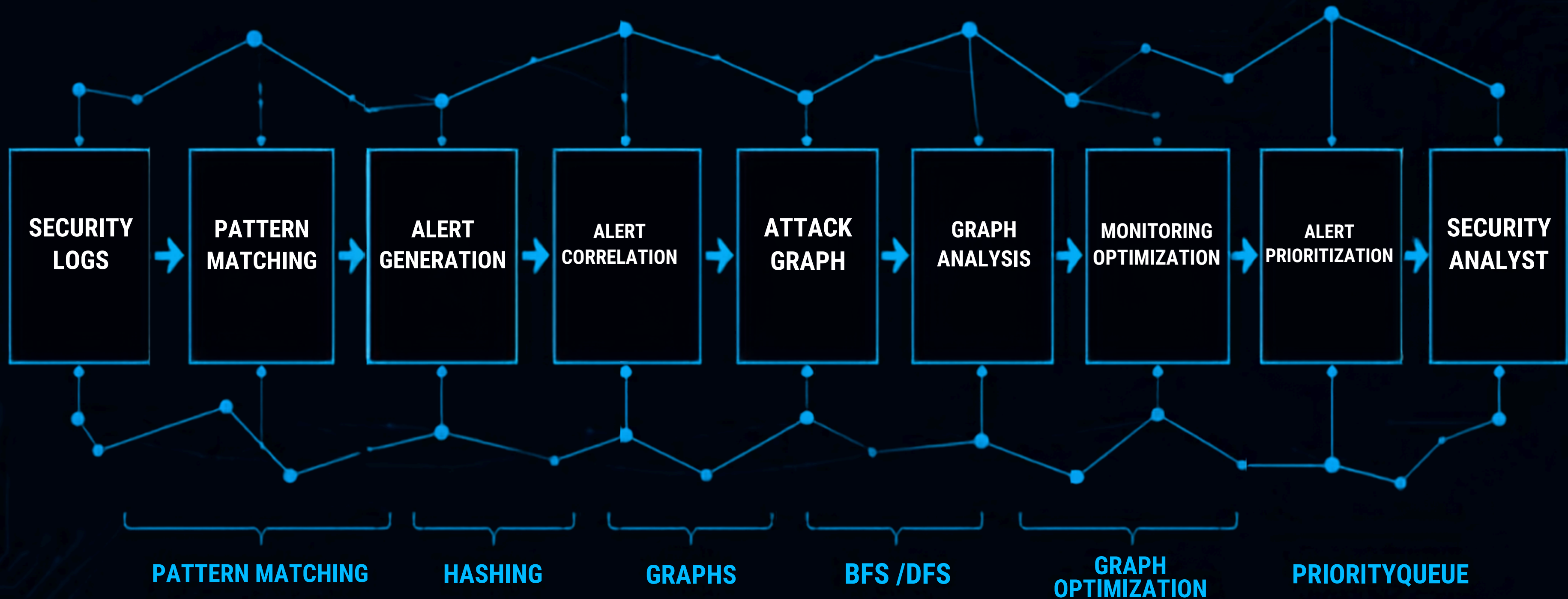
LITERATURE SURVEY

EXISTING APPROACH	LIMITATION	OUR APPROACH
Manual Log Analysis	Time-consuming at scale	Automated pattern matching
Signature-Based Detection	Focuses on individual known events	Alert correlation into kill chains
Rule-Based Alert Systems	Can produce unprioritised alerts	Priority-based alert handling
Basic Alert Grouping	Does not model attack paths	Graph-based attack analysis

KEY OBSERVATION

*Existing approaches address individual parts of threat analysis.
Our project combines these steps using DSA techniques.*

SYSTEM METHODOLOGY



SYSTEM REQUIREMENTS

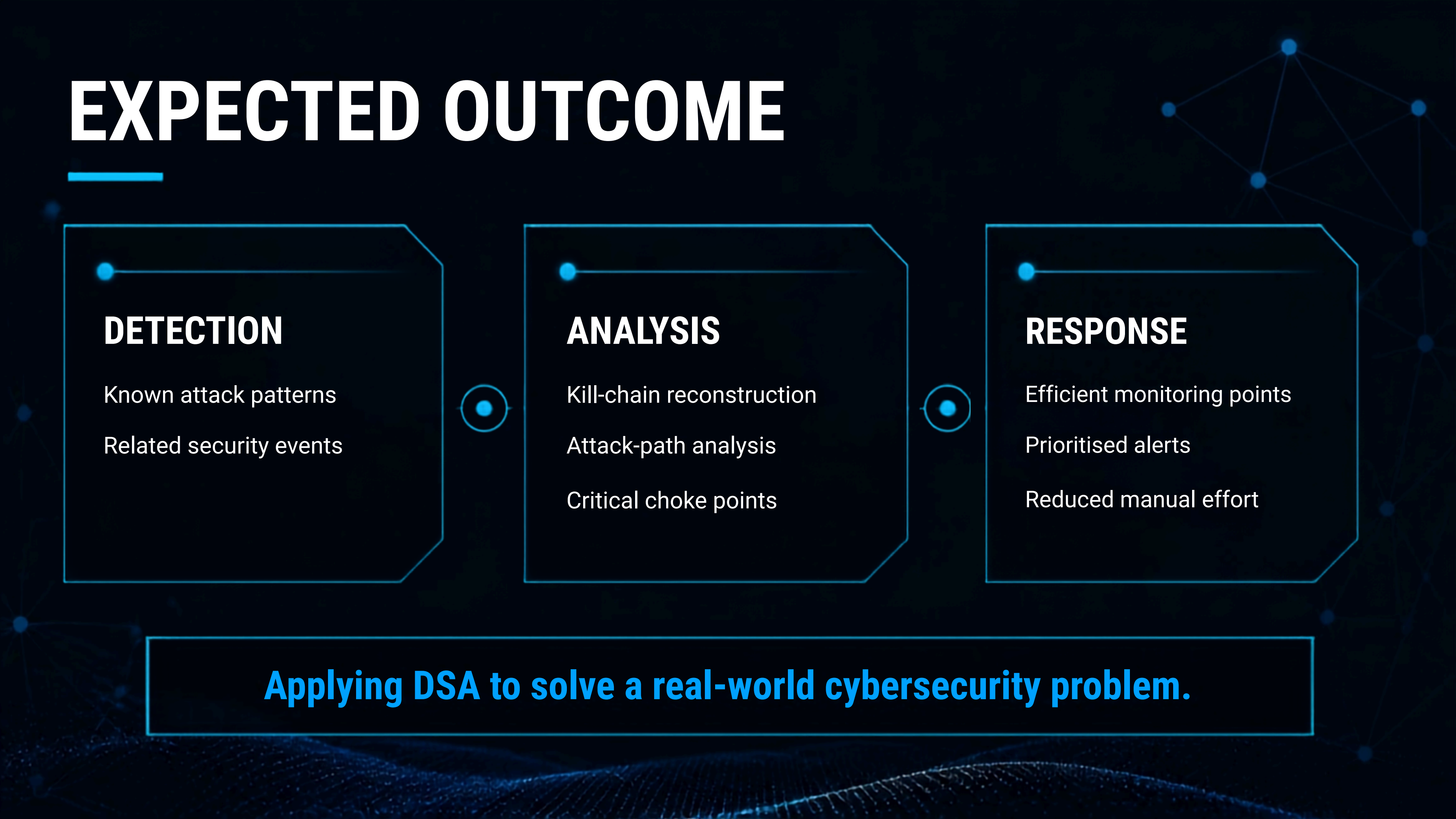
SOFTWARE

- Java (JDK 17+)
- IntelliJ IDEA / Eclipse / VS Code
- Java Collections Framework
- Graph: Adjacency List
- Windows / Linux
- Git / GitHub (*optional*)

HARDWARE

- Intel Core i3 / equivalent or higher
- 4 GB RAM minimum / 8 GB recommended
- 256 GB storage or higher
- Keyboard, mouse & monitor

EXPECTED OUTCOME



DETECTION

- Known attack patterns
- Related security events

ANALYSIS

- Kill-chain reconstruction
- Attack-path analysis
- Critical choke points

RESPONSE

- Efficient monitoring points
- Prioritised alerts
- Reduced manual effort

Applying DSA to solve a real-world cybersecurity problem.